

Dataset : putEMG

Submitted To:
Dr. Raihan Ul Islam
Associate Professor
Department of Computer Science & Engineering

Fall– 2026
Submitted By:
Group-01

Name	Student Id
Mashrur Bin Morshed	2023-2-60-110
Munem Islam Prionto	2023-2-60-024
Kazi Marufin Rahman	2023-2-60-155

Contents

1. Data 1
2. Files and Participants 1
3. One Example Recording 2
4. Noise 2
5. Signal Strength per Gesture 2
6. Channel Correlation 4
7. Another Day and Another Person 5
8. All Recordings 6
9. Conclusions 6

1. Data

The putEMG dataset (Kaczmarek, Mankowski and Tomczynski, Sensors 2019, 19, 3548) contains forearm muscle signals (surface EMG) recorded while people made hand gestures. This report uses the 264 gesture files.

Dataset page: `https://biolab.put.poznan.pl/putemg-dataset/`

Item	Value
Subjects	44 (36 men, 8 women), aged 19 to 37; 39 right-handed, 5 left-handed
Recording days	2 per subject, at least one week apart
Files	264 gesture files (3 trajectories × 2 days × 44 subjects). The 448 force files have no gesture label and are not used.
Electrodes	24 on the right forearm, in 3 bands of 8, 45 degrees apart. `EMG_1` to `EMG_8` elbow band, `EMG_9` to `EMG_16` middle band, `EMG_17` to `EMG_24` wrist band. The first electrode of each band is over the ulna and the numbering runs clockwise.
Recording	5120 Hz, 12-bit, monopolar, total gain 200, hardware filter 3 Hz high-pass and 900 Hz low-pass
Values	raw ADC counts, not filtered or changed in this analysis

Table 1: Overview of the data.

`TRAJ_GT` is the gesture the subject really made, found from video, and is the label used here. `TRAJ_1` is the gesture the subject was asked to make. Label codes: −1 relax (hand free to move), 0 idle (hand still), 1 fist, 2 flexion, 3 extension, 6 pinch thumb-index, 7 pinch thumb-middle, 8 pinch thumb-ring, 9 pinch thumb-small.

2. Files and Participants

Every one of the 44 subjects has exactly 6 gesture files on 2 recording days, with 88 files for each of the trajectories `repeats_long`, `repeats_short` and `sequential`. All 44 subjects appear in the participants table. Most participants are 19 to 23 years old and the forearm diameter at the elbow band is 21 to 30 cm (Figure 1).

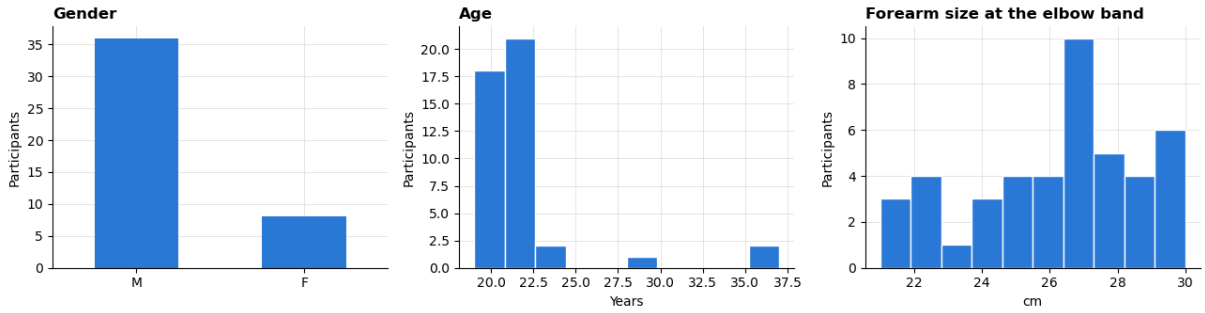


Figure 1: Gender, age and forearm size of the 44 participants.

3. One Example Recording

Sections 3 to 6 use one recording: subject 03, `repeats_long`, day 1. It has 1,746,944 samples (341.2 s) and 32 columns, 24 of them EMG channels. It has no missing values. The values range from −716 to 795 ADC counts. Every channel is centred near zero (mean −1.2 to 0.0) and the standard deviation is 13.8 to 35.2 counts, largest for channels 1 and 9.

Idle takes 182.3 s and relax 42.7 s. Each of the seven gestures takes only 15.5 to 17.5 s (Figure 2). The real gesture equals the requested one in 87.7% of the samples. When a gesture was requested, the real gesture was idle in 7 to 23% of the samples, because people start late and stop early. In Figure 3 the requested gesture (orange) switches on slightly before the real gesture (blue). During each gesture the EMG shows a thick burst, and between gestures it is flat.

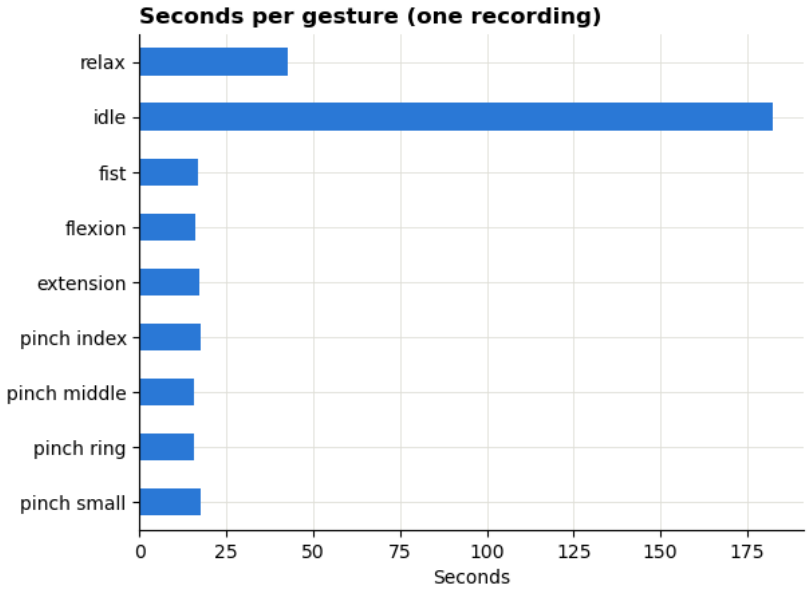


Figure 2: Seconds spent in each label in the example recording.

4. Noise

While idle the muscles are relaxed, so the signal is mostly noise. Its power spectrum has thin spikes at 30, 50 and 60 Hz and many of their multiples (Figure 4). These are interference from the power grid and other equipment, not muscle activity. The signal was not filtered in this analysis, so the interference is included in the results below.

5. Signal Strength per Gesture

Signal strength is measured as the RMS (root mean square) of each channel for each gesture. Each gesture raises a different set of channels (Figures 5 and 6): fist is strongest on channels 1, 2, 9 and 10, extension on channels 6 to 8 and 14 to 16, and flexion is raised on most channels, mostly on the elbow band. Channels 1 and 9 are also the highest while idle, so they are the noisiest. This difference between gestures is what lets a model tell them apart.

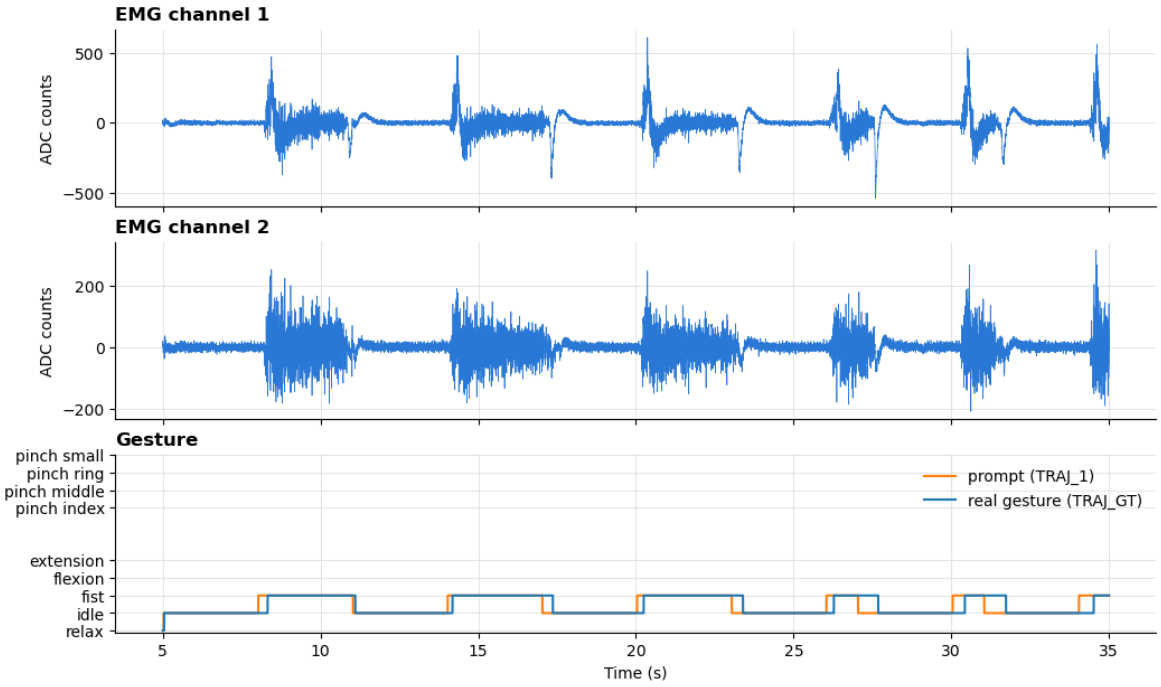


Figure 3: Thirty seconds of the example recording: two EMG channels and the requested and real gesture.

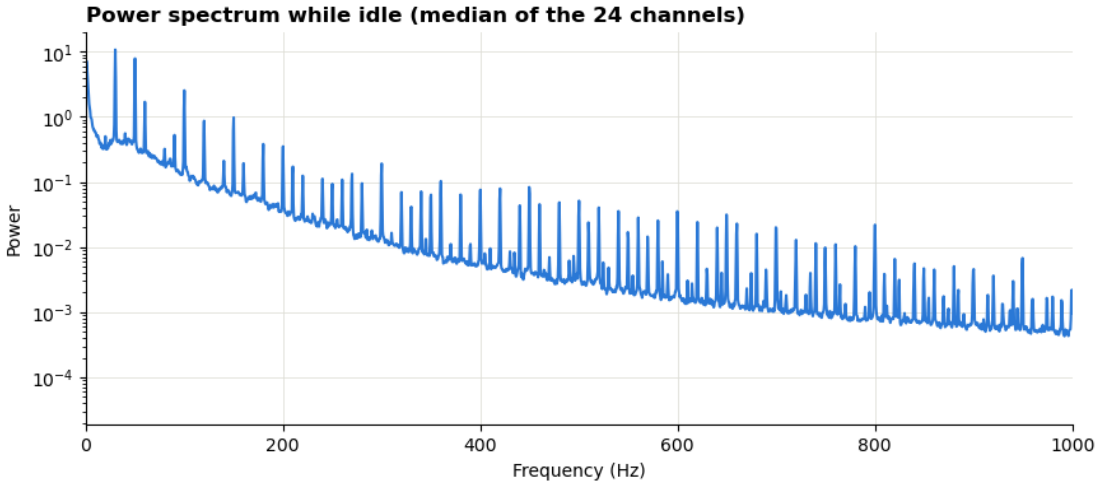


Figure 4: Power spectrum of the idle signal (median of the 24 channels).

Compared with idle, flexion and fist are the strongest gestures and the pinch gestures are the weakest (Table 2). The pinch gestures are only 1.6 to 2.7 times the idle level, which is close to the noise.

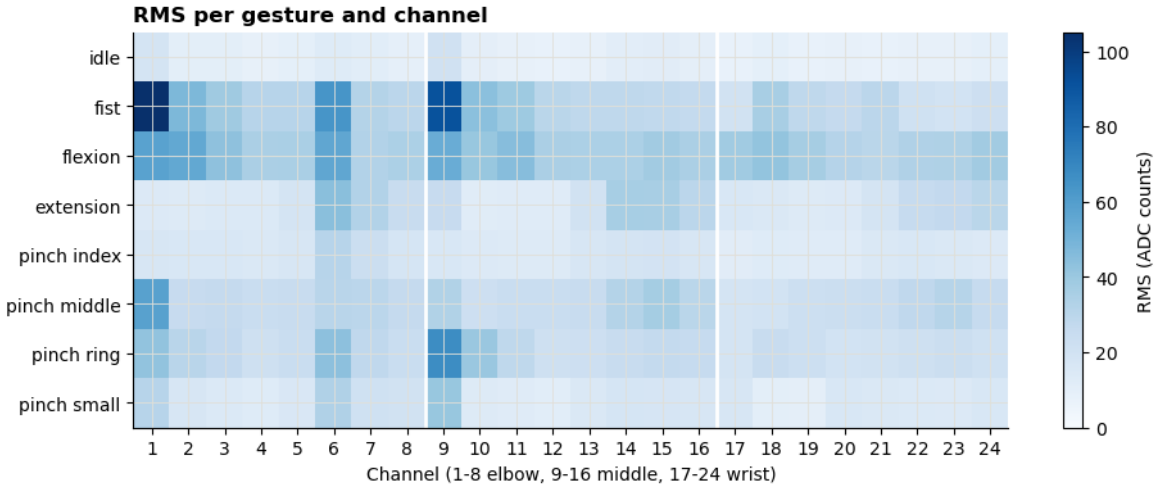


Figure 5: RMS per gesture and channel in the example recording. Darker means stronger.

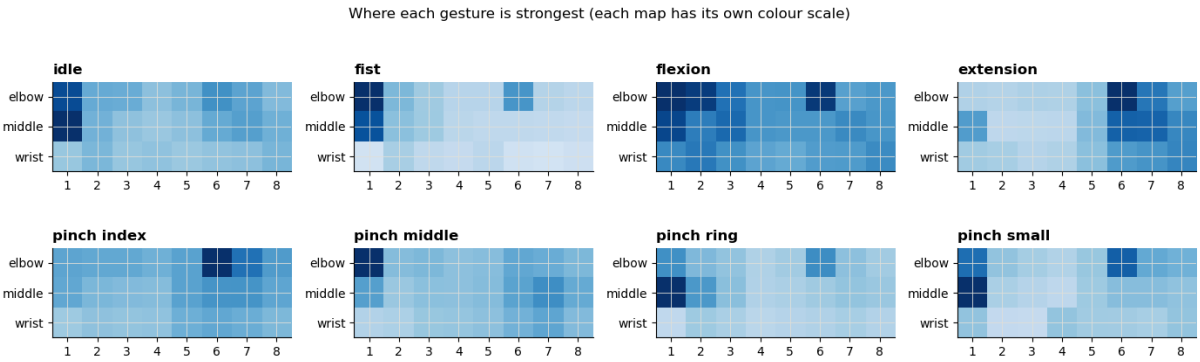


Figure 6: The same values laid out like the electrodes on the arm (3 bands × 8 positions). Each map has its own colour scale.

Gesture	RMS compared with idle
flexion	3.8
fist	3.6
pinch middle	2.7
pinch ring	2.7
extension	2.1
pinch small	1.7
pinch index	1.6

Table 2: Mean RMS of each gesture divided by the mean RMS of idle (raw signal, example recording).

6. Channel Correlation

Channels next to each other are strongly correlated, because neighbouring electrodes sit over the same muscle (Figure 7). The correlation is positive for every pair of channels. Channels 1 and

9 are the weakest links to the others, because slow drift dominates their raw signal (about 80% of their power is below 20 Hz, against 30% for a typical channel).

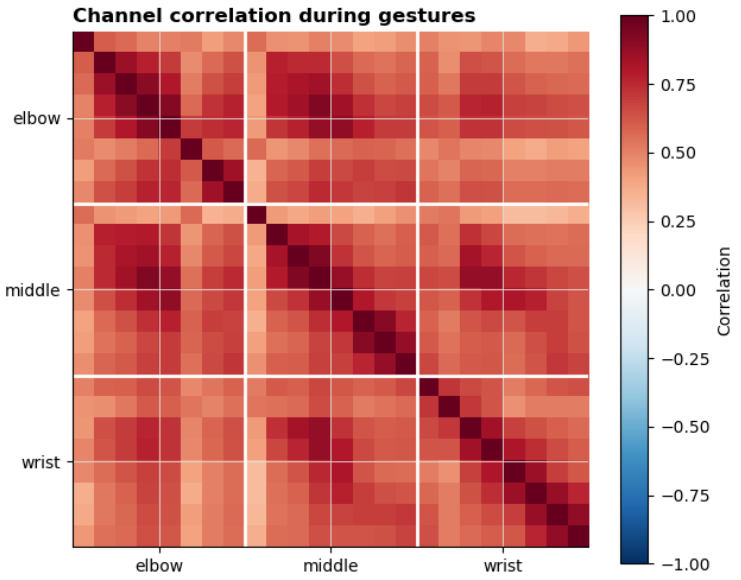


Figure 7: Correlation between the 24 channels during gestures in the example recording.

7. Another Day and Another Person

Electrodes are never put back in exactly the same place, so the pattern changes between days. Figure 8 shows the RMS pattern of subject 03 and subject 04 on both days (`repeats_long`). The four patterns are compared with a similarity from 0 to 1, where 1 means identical and 0 unrelated.

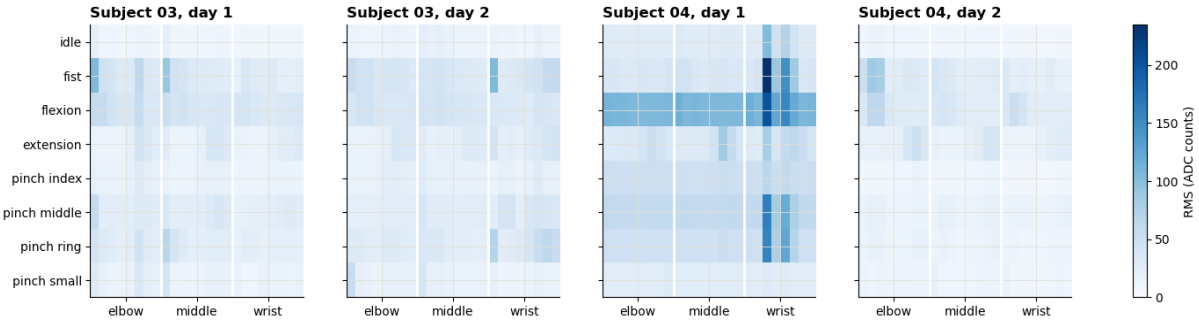


Figure 8: RMS per gesture and channel: subject 03 and subject 04, day 1 and day 2.

The same subject on the other day is 0.53 (subject 03) and 0.30 (subject 04). Different subjects average 0.45, from 0.30 to 0.66. So with these four raw recordings a person is not clearly more alike on another day than another person is. Subject 04 day 1 stands out: flexion is very strong on every channel, and a separate check found 57 samples at the maximum value of the converter in that file. Four recordings are too few to decide.

8. All Recordings

Every one of the 264 recordings was checked for missing values and for time spent in each label. No file has missing values. The data covers 21.4 hours. Idle is 667.7 minutes (52.0%), relax 188.7 minutes (14.7%), and each of the seven gestures only 59.0 to 63.7 minutes (4.6 to 5.0%) (Figure 9). The classes are strongly imbalanced.

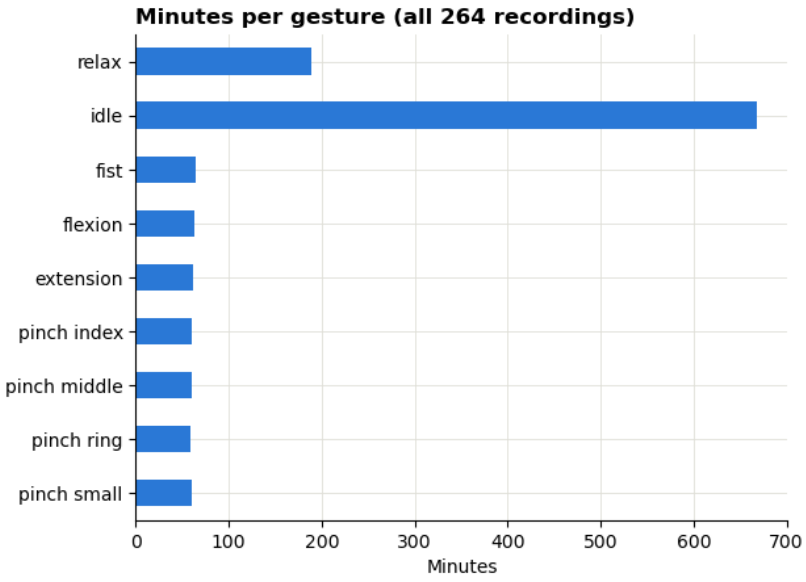


Figure 9: Minutes spent in each label over all 264 recordings.

9. Conclusions

- The data is complete: 264 files, 44 subjects, 2 days each, no missing values.
- Idle is about half of all time, and each gesture is under 5%. Plain accuracy would be misleading for a model.
- Use `TRAJ_GT` as the label. In the example recording it differs from the requested gesture in 12% of the samples, mostly where the real gesture is idle. Drop the relax label (−1).
- The raw signal has interference spikes and noisy channels (1 and 9). Filter it before modelling.
- Gestures differ in their channel patterns, but the pinch gestures are weak (1.6 to 2.7 times idle) and will likely be the hardest to recognise.
- The comparison of days and people is inconclusive with four recordings. Split training and test data by subject, never by random rows or windows.